

# Midterm 2 Review Session

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## Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
```

```
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

## 1 Introduction

This chapter walks through the **most common mistakes** that students made on Midterm 2, which covered survival analysis and Cox proportional hazards models. Each section states a mistake, explains *why* it is wrong, and works through the correct approach.

The exam had two parts:

- **Part 1** (Kaplan-Meier and Nelson-Aalen estimators) used a small data set of survival times: 10, 14, 14+, 15+, 18, 21, 25 months (the + marks censored observations).
- **Part 2** (Cox proportional hazards regression) used a model fit to  $n = 3,142$  men from the Western Collaborative Group Study (WCGS), relating time to incident coronary heart disease (CHD) to baseline cigarette smoking and other covariates.

### Tip

The mistakes below are organized so that the most consequential and most frequent errors come first within each part. If you only have time to review a few things, start at the top of each part.

## 2 Part 1: Kaplan-Meier and Nelson-Aalen

```
library(survival)
library(dplyr)

surv_data <- tibble(
```

```
time = c(10, 14, 14, 15, 18, 21, 25),
death = c(1, 1, 0, 0, 1, 1, 1),
surv = Surv(time, death)
)
```

Here is the full table that everyone was trying to reproduce; we will refer back to it throughout this part.

```
KM_est <- survfit(surv ~ 1, type = "kaplan-meier", data = surv_data)

surv_table <-
  KM_est |>
  summary(censored = TRUE, data.frame = TRUE) |>
  as_tibble() |>
  mutate(
    hazard      = n.event / n.risk,
    nonhazard   = 1 - hazard,
    cusum_hazard = cumsum(hazard),
    surv_KM     = cumprod(nonhazard),
    surv_NA     = exp(-cusum_hazard)
  ) |>
  select(time, n.risk, n.event, hazard, surv_KM, cusum_hazard, surv_NA)

surv_table |> pander::pander()
```

Table 1: Kaplan-Meier and Nelson-Aalen calculations for the Part 1 data

| time | n.risk | n.event | hazard | surv_KM | cusum_hazard | surv_NA |
|------|--------|---------|--------|---------|--------------|---------|
| 10   | 7      | 1       | 0.1429 | 0.8571  | 0.1429       | 0.8669  |
| 14   | 6      | 1       | 0.1667 | 0.7143  | 0.3095       | 0.7338  |
| 15   | 4      | 0       | 0      | 0.7143  | 0.3095       | 0.7338  |
| 18   | 3      | 1       | 0.3333 | 0.4762  | 0.6429       | 0.5258  |
| 21   | 2      | 1       | 0.5    | 0.2381  | 1.143        | 0.3189  |
| 25   | 1      | 1       | 1      | 0       | 2.143        | 0.1173  |

## 2.1 The risk set shrinks by the number of subjects who *leave*

Exam reference: **Exercise 1.1** (6 points) — compute the KM and Nelson-Aalen estimates of  $S(t)$ .

### Common mistake

Reducing the number at risk by **one** at every row, regardless of how many subjects actually left the study. With this mistake, the number at risk after the two censored subjects (at  $t = 14^+$  and  $t = 15^+$ ) was computed as 5 instead of 4.

### Solution

*Solution.* The number at risk  $n_j$  is the number of subjects still under observation *just before* time  $t_j$  — that is, everyone who has neither had the event nor been censored yet.

Track the cohort of 7:

| time | what happens                  | number at risk <i>before</i> this time |
|------|-------------------------------|----------------------------------------|
| 10   | 1 death                       | 7                                      |
| 14   | 1 death <b>and</b> 1 censored | 6                                      |
| 15   | 1 censored                    | 4                                      |

|    |         |   |
|----|---------|---|
| 18 | 1 death | 3 |
| 21 | 1 death | 2 |
| 25 | 1 death | 1 |

At  $t = 14$ , one subject dies *and* a different subject is censored (the 14+), so **two** subjects leave the risk set; the count drops from 6 to 4, not from 6 to 5. Every subject who has an event **or** is censored is removed from all later risk sets.

## 2.2 Nelson-Aalen estimates the *cumulative hazard* first, then exponentiates

Exam reference: **Exercise 1.1** (6 points) — compute the KM and Nelson-Aalen estimates of  $S(t)$ .

### ⚠ Common mistake

Two versions of this error were common:

1. Computing a running sum of arbitrary quantities (e.g. summing survival probabilities, or summing  $1 - \hat{\lambda}_j$ ) instead of summing the hazards  $\hat{\lambda}_j = d_j/n_j$ .
2. Computing the cumulative hazard correctly but then forgetting the final step,  $\hat{S}_{NA}(t) = \exp\{-\hat{\Lambda}(t)\}$ , and reporting the cumulative hazard itself (or the KM product) as the Nelson-Aalen survival estimate.

### Solution

*Solution.* The Nelson-Aalen estimator builds the **cumulative hazard** by adding up the instantaneous hazard contributions at each event time:

$$\hat{\Lambda}(t) = \sum_{j:t_j \leq t} \frac{d_j}{n_j}$$

and then converts it to a survival estimate using  $S(t) = \exp\{-\Lambda(t)\}$ :

$$\hat{S}_{NA}(t) = \exp\{-\hat{\Lambda}(t)\}$$

For example, at  $t = 10$ :

$$\hat{\Lambda}(10) = \frac{1}{7} = 0.143, \quad \hat{S}_{NA}(10) = \exp\{-0.143\} = 0.867$$

Contrast this with Kaplan-Meier, which multiplies conditional survival probabilities:

$$\hat{S}_{KM}(t) = \prod_{j:t_j \leq t} \left(1 - \frac{d_j}{n_j}\right), \quad \hat{S}_{KM}(10) = 1 - \frac{1}{7} = 0.857$$

The two estimators are close but **not** identical, and  $\hat{S}_{NA}(t) \geq \hat{S}_{KM}(t)$  always. Reporting the same numbers for both, or reporting  $\hat{\Lambda}$  where  $\hat{S}_{NA}$  was asked for, loses credit.

## 2.3 Survival estimates are *step functions*: use the most recent event time

Exam reference: **Exercise 1.3** (1 point) — the KM and NA survival estimates for  $t = 17$  months.

### ⚠ Common mistake

For  $\hat{S}(17)$ , looking at the row for  $t = 18$  (the next event *after* 17), or interpolating between rows. Some answers gave only a verbal description (“it’s flat there”) without the numeric value.

### Solution

*Solution.*  $\hat{S}(t)$  is a **right-continuous step function**: it only changes at observed event times and stays flat in between. To evaluate it at  $t = 17$ , find the **most recent event time at or before 17**, which is  $t = 14$  (the subsequent  $t = 15^+$  is a censoring, not an event, so the curve does not step there). So:

$$\hat{S}_{KM}(17) = \hat{S}_{KM}(14) = 0.714, \quad \hat{S}_{NA}(17) = \hat{S}_{NA}(14) = 0.734$$

Give the actual numbers, not just “it’s flat.”

## 2.4 Median survival time is read off the curve, not modeled

*Exam reference: **Exercise 1.4** (1 point) — the KM and NA estimates of median survival time.*

### ⚠ Common mistake

Estimating the median by fitting an exponential model ( $\hat{\lambda} = \text{events}/\text{total follow-up}$ , then  $\hat{E}[T] = 1/\hat{\lambda}$ ). That computes an exponential-model **mean**, not the nonparametric **median**, and answers a different question. A second error was reading the median off the wrong curve or the wrong row.

### Solution

*Solution.* The median survival time is the smallest  $t$  at which the estimated survival curve drops to (or below) 0.5:

$$\hat{t}_{\text{median}} = \min\{t : \hat{S}(t) \leq 0.5\}$$

Reading from Table 1:

- $\hat{S}_{KM}(t)$  first reaches  $\leq 0.5$  at  $t = 18$  (where it drops to 0.476), so the KM median is **18 months**.
- $\hat{S}_{NA}(t)$  first reaches  $\leq 0.5$  at  $t = 21$  (where it drops to 0.319), so the NA median is **21 months**.

We can confirm this in R:

```
quantile(KM_est, p = 0.5)$quantile  
#> 50  
#> 18
```

No distributional assumption (exponential or otherwise) is used: this is a nonparametric read-off of the estimated curve.

## 2.5 How censored times are used: denominator yes, numerator no

*Exam reference: **Exercise 1.5** (2 points) — describe how the censored time(s) are utilized in the Kaplan-Meier estimate.*

### ⚠ Common mistake

Describing only the *symptom* (“the curve doesn’t drop at a censoring time”) without the *mechanism*, or implying that censored subjects are simply dropped from the analysis entirely.

### Solution

*Solution.* A censored subject contributes to the **risk-set denominators** ( $n_j$ ) for every event time up to and including their censoring time, because we know they survived event-free until

then. They **never** contribute to an event **numerator** ( $d_j$ ), because no event was observed for them.

In this example, subjects #3 and #4 (censored at  $14^+$  and  $15^+$ ) are counted in the number at risk at  $t = 10$  and  $t = 14$ , which is why those denominators are 7 and 6. After their censoring times they drop out of all later risk sets. This is exactly what lets Kaplan-Meier use partial information from censored subjects instead of discarding them.

### 3 Part 2: Cox proportional hazards models

For reference, here is the fitted model from the exam. The primary exposure is baseline smoking category (nonsmoker = reference, light smoker  $L = 1$ , heavy smoker  $H = 1$ ), adjusting for age in decades ( $A$ ), behavior pattern ( $P$ ), overweight ( $B$ ), and high cholesterol ( $C$ ).

Table 2: Cox PH regression for the WCGS CHD data

| Characteristic           | $\log(\widehat{HR})$ | $\widehat{HR}$ | 95% CI for HR | $p$     |
|--------------------------|----------------------|----------------|---------------|---------|
| nonsmoker (ref.)         | —                    | —              | —             | —       |
| light smoker ( $L$ )     | 0.36                 | 1.43           | (1.05, 1.95)  | 0.023   |
| heavy smoker ( $H$ )     | 0.78                 | 2.18           | (1.63, 2.91)  | < 0.001 |
| age per decade ( $A$ )   | 0.66                 | 1.94           | (1.56, 2.40)  | < 0.001 |
| behavior type A ( $P$ )  | 0.73                 | 2.06           | (1.58, 2.70)  | < 0.001 |
| overweight ( $B$ )       | 0.30                 | 1.34           | (1.05, 1.72)  | 0.019   |
| high cholesterol ( $C$ ) | 0.75                 | 2.12           | (1.65, 2.71)  | < 0.001 |

The exam also provided the estimated covariance matrix of the coefficient estimates (on the  $\log(HR)$  scale). The entries we need below are:

$$\text{Var}(\hat{\beta}_L) = 0.024840, \quad \text{Var}(\hat{\beta}_H) = 0.021905, \quad \text{Cov}(\hat{\beta}_L, \hat{\beta}_H) = 0.010504$$

#### 3.1 Writing the model: name every assumption and show where it is used

*Exam reference: **Exercise 2.1** (10 points) — write the mathematical form of the proportional hazards model corresponding to Table 1.*

##### Common mistake

Several patterns lost the most points here:

- Treating the question as “interpret the coefficients” and writing only the linear predictor, omitting the **likelihood**, the **distribution functions**, and the **assumptions**.
- Listing assumption *names* without **showing where they are used**, or writing the math without naming the assumption.
- Adding a spurious “**baseline hazard is exponential**” assumption. The Cox model leaves the baseline hazard  $\lambda_0(t)$  **unspecified**; assuming a parametric form is wrong.
- Omitting one or more of the standalone distribution functions (survival, density, hazard, log-hazard, cumulative hazard).

## Solution

*Solution.* A full-credit answer connects the likelihood to the linear predictor through a chain of named components and assumptions. Using  $\tilde{X} = (L, H, A, P, B, C)$ :

**Joint likelihood of the data set:**

$$\mathcal{L} \stackrel{\text{def}}{=} p(\tilde{Y} = \tilde{y}, \tilde{D} = \tilde{d} \mid \mathbf{X} = \mathbf{x})$$

**Marginal likelihood contribution of observation  $i$ :**

$$\mathcal{L}_i \stackrel{\text{def}}{=} p(Y_i = y_i, D_i = d_i \mid \tilde{X}_i = \tilde{x}_i)$$

**Independent-observations assumption** (used to factor the likelihood):

$$\mathcal{L} = \prod_{i=1}^n \mathcal{L}_i$$

**Non-informative censoring assumption** (used so each contribution reduces to survival and hazard terms):  $T_i \perp\!\!\!\perp C_i \mid \tilde{X}_i$ , giving

$$\mathcal{L}_i \propto [f(y_i \mid \tilde{x}_i)]^{d_i} [S(y_i \mid \tilde{x}_i)]^{1-d_i} = S(y_i \mid \tilde{x}_i) \cdot [\lambda(y_i \mid \tilde{x}_i)]^{d_i}$$

**Distribution functions** (define each one):

$$S(t \mid \tilde{x}) \stackrel{\text{def}}{=} P(T > t \mid \tilde{X} = \tilde{x}) = \exp\{-\Lambda(t \mid \tilde{x})\}$$

$$f(t \mid \tilde{x}) \stackrel{\text{def}}{=} \lambda(t \mid \tilde{x}) S(t \mid \tilde{x})$$

$$\lambda(t \mid \tilde{x}) \stackrel{\text{def}}{=} p(T = t \mid T \geq t, \tilde{X} = \tilde{x}) = \frac{f(t \mid \tilde{x})}{S(t \mid \tilde{x})}$$

$$\Lambda(t \mid \tilde{x}) \stackrel{\text{def}}{=} \int_0^t \lambda(u \mid \tilde{x}) du = -\log S(t \mid \tilde{x})$$

$$\eta(t \mid \tilde{x}) \stackrel{\text{def}}{=} \log \lambda(t \mid \tilde{x})$$

**Proportional-hazards assumption** (used to split the hazard into a baseline that depends only on time and a factor that depends only on covariates):

$$\lambda(t \mid \tilde{x}) = \lambda_0(t) \cdot \theta(\tilde{x})$$

where  $\lambda_0(t)$  is the **unspecified** baseline hazard.

**Logarithmic-link assumption** (used to make the covariate factor a function of a linear predictor):

$$\eta(t \mid \tilde{x}) = \eta_0(t) + \Delta\eta(\tilde{x}), \quad \theta(\tilde{x}) = \exp\{\Delta\eta(\tilde{x})\}$$

**Linear functional-form assumption** (used to write the covariate term as a linear combination):

$$\Delta\eta(\tilde{x}) = \tilde{x} \cdot \tilde{\beta} = \beta_L l + \beta_H h + \beta_A a + \beta_P p + \beta_B b + \beta_C c$$

Notice that the baseline hazard  $\lambda_0(t)$  is carried along symbolically the whole time — we never assume a shape for it.

## 3.2 Interpreting a hazard ratio: magnitude, reference, adjustment, and significance

*Exam reference: **Exercise 2.2** (4 points) — summarize how baseline smoking category is associated with hazard of incident CHD.*

 Common mistake

- Reporting the direction of the effect but omitting the **magnitude** (“X% higher/lower hazard”) or the **reference group**.
- Forgetting to say the estimate is **adjusted for** / **holding constant** the other covariates.
- Omitting **statistical significance** (whether the 95% CI excludes 1, or  $p < 0.05$ ).
- Calling the hazard ratio a **risk** or an **odds ratio**. It is a ratio of *hazards*, not of risks or odds.

Solution

*Solution.* A complete interpretation has four parts: magnitude, reference group, adjustment, and significance. For the smoking effect in Table 2:

Adjusting for age, behavior pattern, overweight, and cholesterol, **light smokers** had an estimated hazard of incident CHD about **43% higher** than **nonsmokers** ( $\widehat{HR} = 1.43$ ); this is statistically significant at the 0.05 level, since the 95% CI (1.05, 1.95) excludes 1 ( $p = 0.023$ ).

**Heavy smokers** had an estimated hazard about **118% higher** (roughly double;  $\widehat{HR} = 2.18$ ) than nonsmokers, also adjusting for the other covariates; this is highly significant ( $p < 0.001$ , CI (1.63, 2.91)).

Note “43% higher” comes from  $1.43 - 1 = 0.43$ , and “118% higher” from  $2.18 - 1 = 1.18$ . Interpret each smoking level **separately** against the reference, rather than lumping them together.

### 3.3 A hazard ratio for a non-unit change: exponentiate

*Exam reference: Exercise 2.3 (2 points)* — the hazard ratio associated with a 7.5-year increase in age.

 Common mistake

For the hazard ratio associated with a **7.5-year** increase in age (with the table reporting the HR per **decade**), the errors were:

- Multiplying:  $1.94 \times 0.75 = 1.46$ . **Wrong** — the HR for a multi-unit change is *not* linear in the HR.
- Arithmetic slips that produced answers like 2.08.

Solution

*Solution.* On the **log** scale, the effect is linear: a change of  $c$  units multiplies the log-hazard by  $c \cdot \beta$ . So the hazard ratio for a  $c$ -unit change is

$$HR(c) = \exp\{c\beta\} = (\exp\{\beta\})^c = HR^c$$

A 7.5-year increase is  $c = 0.75$  decades, and the per-decade  $\widehat{HR} = 1.94$  (i.e.  $\hat{\beta}_A = \log 1.94 = 0.66$ ):

$$\widehat{HR}(0.75) = 1.94^{0.75} = \exp\{0.75 \times 0.66\} = \exp\{0.495\} = 1.64$$

So a 7.5-year-old man has about **64% higher** estimated hazard of CHD, all else equal. The operation is **exponentiation** ( $HR^{0.75}$ ), not multiplication.

### 3.4 Comparing two coefficients: variance of a difference

*Exam reference: Exercise 2.4 (2 points)* — test whether the hazard of incident CHD differs between heavy and light smokers; compute the  $z$ -statistic and two-sided  $p$ -value.



 Common mistake

This was the **single most consequential Part 2 error**. To test  $H_0 : \beta_H = \beta_L$ , the standard error of  $\hat{\beta}_H - \hat{\beta}_L$  was computed incorrectly in several ways:

- Using a **single covariance entry**  $\text{Cov}(\hat{\beta}_L, \hat{\beta}_H)$  as the variance.
- **Adding** 2 Cov instead of **subtracting** it.
- Plugging in the **hazard ratios** (2.18, 1.43) instead of the **coefficients** (0.78, 0.36) in the numerator.
- Guessing a value for  $z$  because the variance-of-a-difference formula was not on the formula sheet.

Solution

*Solution.* Work on the  $\log(HR)$  (coefficient) scale, where the estimates are approximately normal. The point estimate of the difference is

$$\hat{\Delta} = \hat{\beta}_H - \hat{\beta}_L = 0.78 - 0.36 = 0.42$$

The variance of a **difference** of two estimates uses **all three** relevant entries of the covariance matrix:

$$\text{Var}(\hat{\beta}_H - \hat{\beta}_L) = \text{Var}(\hat{\beta}_H) + \text{Var}(\hat{\beta}_L) - 2 \text{Cov}(\hat{\beta}_H, \hat{\beta}_L)$$

The  $-2$  Cov term is essential — its sign is **minus** for a difference. Plugging in:

```
var_H <- 0.021905
var_L <- 0.024840
cov_HL <- 0.010504

diff <- 0.78 - 0.36
var_diff <- var_H + var_L - 2 * cov_HL
se_diff <- sqrt(var_diff)

z <- diff / se_diff
p_value <- 2 * pnorm(-abs(z))

c(diff = diff, se = se_diff, z = z, p_value = p_value) |> round(4)
#>      diff      se      z p_value
#> 0.4200 0.1604 2.6180 0.0088
```

So  $z = 0.42/0.160 = 2.62$ , two-sided  $p = 0.0088$ . We **reject**  $H_0$ : the hazard of CHD differs significantly between heavy and light smokers, holding the other covariates constant.

**Note**

Compare with the **naive** standard error that ignores the covariance,  $\sqrt{\text{Var}(\hat{\beta}_H) + \text{Var}(\hat{\beta}_L)} = 0.216$ . Because  $\hat{\beta}_H$  and  $\hat{\beta}_L$  are *positively* correlated, subtracting 2 Cov makes the correct SE (0.160) **smaller**, which sharpens the test. Using a single covariance cell as the variance gave a wildly wrong SE and an absurd  $z$ .

### 3.5 A confidence interval for a hazard ratio: build it on the log scale, then exponentiate

*Exam reference: Exercise 2.5 (2 points)* — a 95% confidence interval for the hazard ratio comparing heavy to light smokers.

#### ⚠ Common mistake

- Forming the interval directly on the HR scale,  $\widehat{HR} \pm 1.96 \cdot SE$  (e.g.  $1.52 \pm 0.31$ ), instead of on the log scale.
- Building the log-scale interval correctly but **never exponentiating**, then reporting a log-scale interval as if it were the HR (sometimes even declaring “significant” while the reported interval contained 1).
- Reusing the wrong SE from the Wald-test mistake above.

#### Solution

*Solution.* A hazard ratio is positive and its sampling distribution is skewed, so the symmetric normal approximation is applied to  $\log(HR)$ , and the endpoints are **then** exponentiated. For the heavy-vs-light comparison ( $\widehat{\Delta} = 0.42$ ,  $SE = 0.160$ ):

$$\text{log-scale CI: } 0.42 \pm 1.96 \times 0.160 = (0.106, 0.734)$$

$$\text{HR CI: } (e^{0.106}, e^{0.734}) = (1.11, 2.08)$$

```
est <- 0.42
se <- sqrt(0.021905 + 0.024840 - 2 * 0.010504)
exp(est + c(-1, 1) * 1.96 * se) |> round(2)
#> [1] 1.11 2.08
```

The point estimate is  $\widehat{HR} = e^{0.42} = 1.52$ , and the 95% CI (1.11, 2.08) **excludes 1**, consistent with the significant Wald test in Section 3.4. Always exponentiate the endpoints, and check that your CI and your hypothesis test agree.

## 4 Summary of the most common errors

| # | Exam Q | Topic                                    | The fix                                                                                    |
|---|--------|------------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | 1.1    | Risk set after censoring (Section 2.1)   | Remove <b>everyone</b> who has an event <i>or</i> is censored from later risk sets.        |
| 2 | 1.1    | Nelson-Aalen (Section 2.2)               | Sum hazards $d_j/n_j$ , then $\hat{S}_{NA} = \exp\{-\hat{\Lambda}\}$ .                     |
| 3 | 1.3    | Reading $\hat{S}(t)$ (Section 2.3)       | Step function: use the most recent event time $\leq t$ ; give the number.                  |
| 4 | 1.4    | Median survival (Section 2.4)            | First $t$ with $\hat{S}(t) \leq 0.5$ — not a mean, not an exponential fit.                 |
| 5 | 1.5    | Role of censoring (Section 2.5)          | Denominator until censoring; never the event numerator.                                    |
| 6 | 2.1    | Writing the model (Section 3.1)          | Name <b>and</b> apply every assumption; baseline hazard stays unspecified.                 |
| 7 | 2.2    | Interpreting an HR (Section 3.2)         | Magnitude + reference + adjustment + significance; HR $\neq$ risk/odds.                    |
| 8 | 2.3    | HR for a non-unit change (Section 3.3)   | $HR^c = \exp\{c\beta\}$ — exponentiate, don't multiply.                                    |
| 9 | 2.4    | Comparing two coefficients (Section 3.4) | $\text{Var}(\hat{\beta}_H - \hat{\beta}_L) = \text{Var}_H + \text{Var}_L - 2 \text{Cov}$ . |

| #  | Exam Q | Topic                                          | The fix                                                     |
|----|--------|------------------------------------------------|-------------------------------------------------------------|
| 10 | 2.5    | CI for an HR<br>(Section <a href="#">3.5</a> ) | Build on the log scale, then<br>exponentiate the endpoints. |